STOCHASTIC METHODS IN FINANCE
STAT0013
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Exercises Workshop 3: Itô's Formula

Unless otherwise specified, B_t below is a (standard) Brownian Motion.

1. Assume $X_t = at + \sigma B_t$.
   (a) Find the SDE corresponding to e^{-X_t}.
   (b) Find the SDE corresponding to $\cos(e^{X_t})$.

2. Assume that S_t satisfies $dS_t = (a - \delta)S_t dt + \sigma S_t dB_t$. Find the SDE corresponding to
   (a) $X_t = \log(S_t)$
   (b) $X_t = \sqrt{S_t}$.
   (c) $X_t = e^{t/2}\sin(S_t)$.

3. (a) Assume that the price process S_t for an underlying asset follows the stochastic differential equation (SDE) $dS_t = \mu S_t dt + \sigma S_t dB_t$. Let $F(S_t, t)$ be the price of a forward contract agreed at time t for delivery of an underlying asset at time $T > t$. Show that the SDE for $F(S_t, t)$ for $t < T$ is $dF_t = (\mu - r)F_t dt + \sigma F_t dB_t$.
   (b) At time t, what is the probability distribution for $F(S_{t_1}, t_1)$, where $t < t_1 < T$? Explain why, and give all relevant parameters.

4. Assume that we have $d\sqrt{V_t} = (2 - 4\sqrt{V_t})dt + V_t dB_t$. Find the SDE equation satisfied by V_t.

5. Assume that S_t is standard Brownian Motion. Let X_t satisfy the SDE

$$dX_t = 3S_t dt + 3S_t^2 dB_t,$$

   with $X_0 = 0$. Find X_t explicitly.

6. Assume that S_t is standard Brownian Motion. Let X_t satisfy the SDE

$$dX_t = e^{S_t}\cos(S_t)dt + e^{S_t}[\sin(S_t) + \cos(S_t)]dB_t,$$

with $X_0 = 5$. Find X_t explicitly.

7. Suppose a stochastic process S_t follows the stochastic differential equation $dS_t = \mu_t dt + \sigma_t dB_t$. Suppose that $S_0 = 1$, $\mu_t = 1$, $\sigma_t = 2$ for $0 < t < 3$, and that $\mu_t = 5$, $\sigma_t = 1$ for $t > 3$. What is the probability distribution of S_5? Explain your working.

8. Suppose that a process S_t follows the stochastic differential equation $dS_t = \mu S_t dt + \sigma S_t dB_t$, where $S_0 = k$ and μ, and σ are positive constants. Derive the stochastic differential equation for the process Y_t, where $Y_t = S_t^3$. Use this equation to determine the probability distribution for Y_t, for any $t > 0$.

9. Assume that a stock price process, S_t, follows the stochastic differential equation $dS_t = \mu S_t dt + \sigma S_t dB_t$. Consider at time 0 a unit investment on a stock and a unit investment on the riskless asset. Find an expression for the probability that the investment on the stock will be worth more than the investment on the riskless asset at time $T > 0$. Assume that the risk-free rate is r.
   Explain how an investor could use a put option with strike $S_0 e^{rT}$ and maturity T to cover her losses in the event that the stock fails to outperform the riskless asset.